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SECTION 1: HYDRAULIC STABILITY LIMITS & FLOODING CRITERIA

1.1 Allowable Differential Pressure Profile:

- Overall Column Differential Pressure (Bottom - Overhead): ≤ 0.65 bar.
- Single Tray Differential Pressure: 8.0 to 14.0 mbar (0.008 to 0.014 bar).
- Jet Flooding Margin: Operating vapor velocity must not exceed 82% of flooding velocity.

SECTION 2: TEMPERATURE & TOP PRODUCT CONTROL WINDOW

2.1 Overhead Vapor Temperature: 112 deg C to 122 deg C. High Alarm: 128 deg C.

2.2 Top Flash Pressure: 1.45 to 1.65 bar absolute.

2.3 Flash Zone Operating Temperature: 360 deg C to 368 deg C.

SECTION 3: WEAPONS AGAINST FOULING, WEAPING & CORROSION

3.1 Desalted Crude Salt Content Limit: Maximum 3.0 PTB (Pounds per Thousand Barrels).

3.2 Overhead pH: Maintain neutralizer injection to sustain water condensate pH 6.0 - 6.8.

3.3 Downcomer Clearances: Minimum downcomer clearance of 38 mm on heavy wash trays.

SECTION 4: EMERGENCY MITIGATION PROTOCOL

Condition: Column DP > 0.75 bar + Liquid Carryover.

Immediate Action: Reduce stripping steam flow by 20%, increase heavy naphtha reflux.